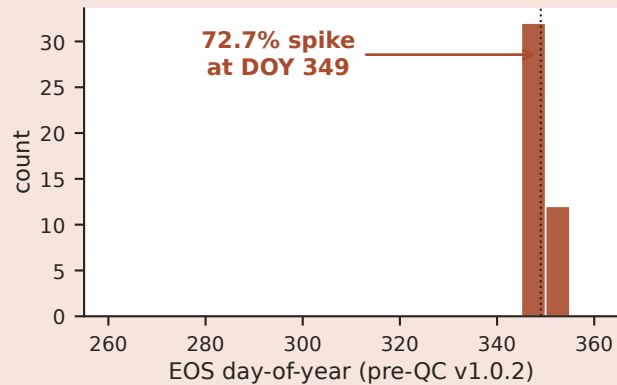


Quantisation artefacts in Sentinel-2 rice phenology:

a reproducible QC framework for cyclone-impact studies

A three-gate QC framework collapses boundary spikes from 72.7% → 8.3% and turns false-positive cyclone effects into a robust null DiD.

1. The problem



Standard double-logistic fits concentrate 72.7% of EOS and 65.6% of POS estimates on single boundary DOYs — artefacts that mimic genuine cyclone signals.

2. Three-gate QC

Gate A Mode-share ≤ 0.20

Boundary-spike test

Gate B Biological plausibility

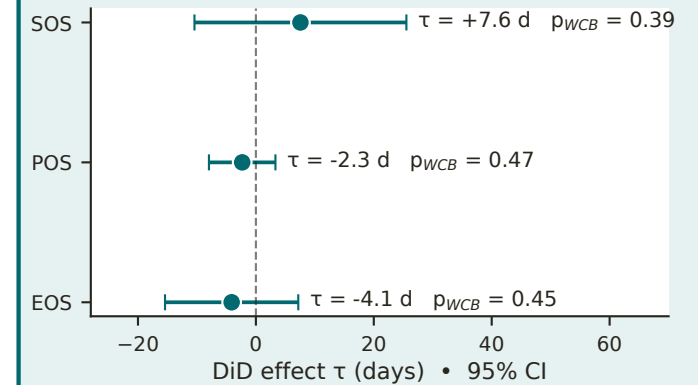
DOY within agronomic range

Gate C Fit quality

$R^2 \geq 0.70$, $RMSE \leq 0.10$

Each panel observation must pass all three gates before entering the causal panel.

3. Robust null



After QC, all three phenometrics return null ($p_{WCB} > 0.38$). The QC framework eliminates the false positives the artefact created.